# Smooth attainment and a genuine bubbling sublevel on a Godeaux blowup

Research continuation

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**Abstract**

The existence target is a smooth global minimizer of $P(g) = \int |\operatorname{Ric}_g^\circ|^2$ which is not Bach-flat. On the fixed classical Godeaux blowup, the supplied strict Burns insertion theorem gives $p_* < 8\pi^2$. We exhibit an actual Eguchi–Hanson degeneration on this same smooth manifold at a level strictly below $8\pi^2$. Thus this bound is not a compactness threshold. We also give one aggregate characteristic class estimate for marked negative-definite splittings, identifying precisely why it does not detect this degeneration. These conclusions do not establish attainment or nonattainment of $p_*$.

## 1 The existence question

Let $X$ be the quotient of the smooth Fermat quintic by the free action

$$\gamma[x_1 : x_2 : x_3 : x_4] = [\zeta x_1 : \zeta^2 x_2 : \zeta^3 x_3 : \zeta^4 x_4], \qquad \zeta^5 = 1,$$

where $\zeta$ is primitive, and put $M = X\#\overline{\mathbb{CP}}^2$. Then $K_X^2 = 1$, $\chi(M) = 12$, $\tau(M) = -8$, and $T = 2\chi + 3\tau = 0$. The inherited classical monopole estimates and the project's strict negative Kähler–Einstein insertion theorem give

$$\frac{16\pi^2}{3} \le p_*(M) := \inf_{g \in \operatorname{Met}(M)} P(g) < 8\pi^2. \tag{1}$$

Here the strict upper bound is an input from the supplied manuscript, not independently re-proved in this note. The exact input is its general KE expansion, at one insertion point,

$$\frac{P(g_\varepsilon)}{8\pi^2} = 1 - \frac{s^2}{64}\varepsilon^2 \log(1/\varepsilon) + O(\varepsilon^2).$$

It is supported in a regular coordinate ball and keeps the outside metric unchanged; see the source record at the end.

A smooth minimizer would have constant scalar curvature by conformal variation. All conformal classes here have negative Yamabe constant by the actual monopole input. A Bach-flat $P$-critical metric of nonzero constant scalar curvature is Einstein, by the two Euler–Lagrange equations. But Einstein and $T = 0$ imply $s = W^+ = 0$ by Chern–Weil, contradicting the scalar estimate $\int s^2 \ge 32\pi^2$. Thus smooth attainment on this $M$ would solve the desired non-Bach-flat minimizer problem.

The inference missing from (1) is smooth attainment. Anderson's ideal-minimizer theorem [1] admits orbifold and open/collapsing alternatives; it does not remove them from (1).

## 2 An explicit nodal Godeaux degeneration

Define the invariant quintic

$$\begin{aligned} F = {} & 2(x_1^5 + x_2^5 + x_3^5 + x_4^5) \\ & - x_1^3 x_3 x_4 - x_1^2 x_2^2 x_4 - x_1^2 x_2 x_3^2 - x_1 x_2^3 x_3 \\ & - x_1 x_2 x_4^3 - x_1 x_3^2 x_4^2 - x_2^2 x_3 x_4^2 - x_2 x_3^3 x_4. \end{aligned}$$

Every monomial has weight zero modulo five. At $[1:1:1:1]$ the gradient vanishes and the homogeneous Hessian is

$$40 I_4 - 10 \mathbf{1}\mathbf{1}^{\mathsf{T}}.$$

Its only kernel is the radial line, so the projective singularity is a node. For completeness, exact polynomial reduction gives, in the chart $x_4 = 1$,

$$\left( \partial_{x_i} F \big|_{x_4=1} : 1 \le i \le 4 \right) = (x_1 - x_3^3,\ x_2 - x_3^2,\ x_3^5 - 1) \quad \text{in } \mathbb{Q}[x_1, x_2, x_3].$$

On $x_4 = 0$, in each of the three charts $x_i = 1$, the derivative ideal is the unit ideal. Hence the hypersurface has precisely five nodes, a single free $\mathbb{Z}_5$ orbit. The derivative-ideal calculation and the Hessian were verified by exact rational arithmetic in the accompanying script.

The only fixed points of nonidentity group elements in projective space are the coordinate vertices; none lies on $F = 0$. Therefore

$$Z_0 = \{F = 0\}/\mathbb{Z}_5$$

has exactly one $A_1$ singularity. Its canonical bundle is ample: the quotient pullback is $\mathcal{O}(1)$ by adjunction.

The invariant pencil $F + u \sum x_i^5$ smooths the nodes for generic small $u \neq 0$. The smooth invariant quintics avoiding the fixed vertices form a connected complex Zariski-open set. Equivariant Ehresmann triviality identifies their quotients with the same oriented smooth $X$. The $A_1$ Milnor fibre is diffeomorphic to the disk bundle of degree $-2$; equivalently, its smoothing and minimal resolution have the same oriented smooth filling. Consequently $X$ contains a smooth sphere $C$ with

$$C^2 = -2, \qquad K_X \cdot C = 0.$$

After a blowup outside this region, $E \cdot C = 0$ as well. This is an embedded sphere, rather than just a root of the intersection lattice.

## 3 Bubbling strictly below the known upper threshold

**Proposition 1.** *Assume the supplied local strict Burns insertion theorem stated in Section 1. There are smooth metrics $g_\lambda$ on the fixed manifold $M$ such that*

$$P(g_\lambda) \longrightarrow q < 8\pi^2,$$

*and a rescaled region converges to the nonflat Ricci-flat Eguchi–Hanson metric. After volume normalization this sequence has no smooth convergent subsequence modulo diffeomorphisms.*

*Proof.* The ample canonical orbifold $Z_0$ admits a negative KE orbifold metric $h_0$ by the orbifold Aubin–Yau theorem; see [2, 3]. Apply the supplied insertion at a regular point away from the node. Fix its parameter sufficiently small to obtain an orbifold metric $h$ on the one-point blowup of $Z_0$ with $P(h) = q < 8\pi^2$. It equals $h_0$ near the node.

Replace the node with Eguchi–Hanson at length scale $\lambda$. In shared end coordinates the differentiated expansions are

$$h = \delta + O(r^2), \qquad g_{\mathrm{EH},\lambda} = \delta + O(\lambda^4 r^{-4}).$$

Interpolate at $\rho = \lambda^{1/2}$. The annulus has volume $O(\rho^4)$ and curvature bounded by $C(1+\lambda^4\rho^{-6})$, so

$$\int_{\text{annulus}} |\operatorname{Ric}^\circ|^2 \le C(\rho^4 + \lambda^8\rho^{-8}) \longrightarrow 0.$$

The removed Einstein orbifold ball and the inserted Ricci-flat interior both have $P = 0$. Thus $P(g_\lambda) \to q$. The resulting smooth manifold is the smoothing of $Z_0$ blown up at a separate point, hence is $M$. The total volume tends to the positive volume of $h$, whereas the maximum Riemann curvature in the EH region grows as $\lambda^{-2}$. Uniform volume normalization preserves this curvature divergence. □

This proposition makes no assertion that $q = p_*(M)$. It disproves the suggested compactness of the whole sublevel below $8\pi^2$, not smooth attainment itself. Its ALE interpolation is classical; the supplied strict Burns saving is what puts the sequence below that threshold.

## 4 What the actual classes control in an entire splitting

The following estimate collects all marked components simultaneously. It is conditional on an appropriate splitting and actual field passage; it is not a claim that every minimizing sequence already supplies them.

**Proposition 2.** *Suppose a bulk $B$ and finitely many ALE pieces $N_i$ give a marked rational orthogonal decomposition, with negative-definite core subspaces. Assume relative Chern–Weil additivity $T_B + \sum_i T_i = 0$ and actual bulk monopole input for the restrictions of $K + E$ and $K - E$, sufficient for the classical mixed curvature estimate. Write*

$$K = K_B + \sum_i K_i, \qquad E = E_B + \sum_i E_i, \qquad d = -\sum_i K_i^2, \quad e = -\sum_i E_i^2.$$

*With the standard operator norm for Weyl curvature, put*

$$C_i = \int_{N_i} \left(4|W^+|^2 + \frac{s^2}{12}\right) \ge 0.$$

*Then the total energy of the configuration satisfies*

$$P_B + \sum_i P_i \ge \frac{16\pi^2}{3}\big(1 + d + \max\{e-1, 0\}\big) + \sum_i C_i. \tag{2}$$

*Proof.* Since $K^2 = 1$ and $E^2 = -1$, one has $K_B^2 = 1 + d$ and $E_B^2 = e - 1$. The average of the two self-dual projection squares is

$$\frac{|(K_B + E_B)^+|^2 + |(K_B - E_B)^+|^2}{2} = |K_B^+|^2 + |E_B^+|^2 \ge 1 + d + \max\{e-1, 0\}.$$

Apply the classical inequality $P_B \ge 8\pi^2(\frac{2}{3}\max_\pm |(K_B \pm E_B)^+|^2 - T_B)$. For each core, relative Chern–Weil gives $P_i = C_i - 8\pi^2 T_i$. Addition cancels all the relative $T$ terms and yields (2). □

In particular, configurations of total energy less than $8\pi^2$ satisfy

$$d + \max\{e-1, 0\} + \frac{3}{16\pi^2}\sum_i C_i < \frac{1}{2}.$$

For the actual $A_1$ sphere above, both restrictions vanish and the Ricci-flat ASD core has $C_i = 0$. The aggregate estimate thus retains only $16\pi^2/3$. The missing information cannot be supplied by counting that sphere again.

For comparison, $r$ disjoint degree-$-3$ cores with canonical degree one give $d = r/3$, excluding $r \geq 2$ at this energy. A chain $[-3, -2, \ldots, -2]$ of rank $r$ with canonical degrees $(1, 0, \ldots, 0)$ instead has $d = r/(2r+1) < 1/2$, from its inverse intersection matrix. This last calculation does not assert that every such chain embeds in $M$.

## 5 Attainment does not require compactness of every minimizing sequence

There is also a concrete example within the construction above. On the unblown smooth $X$, a negative KE metric exists and attains $\inf P = 0$. Resolving the node of $(Z_0, h_0)$ by shrinking EH metrics, without making the Burns insertion, gives another sequence on $X$ with $P \to 0$ and unbounded curvature. Thus a smooth global minimizer and a degenerating minimizing sequence coexist on the same smooth manifold.

Accordingly, the existence target for $M$ is to obtain at least one smooth global minimizer. Strict comparison with all ideal boundary configurations would be one sufficient route, if their compactification and regularity were established; it is not a necessary extra theorem. A suitable selection of minimizing sequences or a smooth minimizing construction could also succeed. Proposition 1 does not decide among these routes.

What is unresolved is a genuinely minimizing comparison: no result here shows that a smooth competitor is lower than every relevant optimized singular configuration, and no smooth metric is shown to attain $p_*(M)$. The previous single-core mass calculations therefore remain tools for one stratum, rather than a proof of Q3.

**Provenance of the nonclassical input.** The strict insertion expansion is taken from the supplied `01_Research_Synthesis.tex`, subsection "The global upper bound that changed during the night", and its referenced A02.59D source. That subsection already extends the construction to canonical models with ADE points. The explicit nodal quintic, its exact finite verification, and the aggregate calculation above explain the consequence for this fixed candidate. They are not assertions of a newly proved general attainment theorem or an exhaustive originality claim.

## References


[1] M. T. Anderson, *Canonical metrics on 3-manifolds and 4-manifolds*, especially Theorem 4.10 and Remark 4.12, arXiv:math/0511034.

[2] R. Kobayashi, *Einstein–Kähler V-metrics on open Satake V-surfaces with isolated quotient singularities*, Math. Ann. **272** (1985), 385–398, doi:10.1007/BF01455566.

[3] I. Şuvaina, *The Yamabe invariant of a class of symplectic manifolds*, Proposition 2.4 and proof of Theorem 3.4, arXiv:1410.1236.

[4] C. LeBrun, *Ricci curvature, minimal volumes, and Seiberg–Witten theory*, arXiv:math/0003068.